Project Idea

Nowadays, AI tools like chatGPT and Gemini answers quickly, but sometimes their answers can be:

- Incorrect → the answer contains wrong facts
- Irrelevant → it doesn't answer the question
- Hallucinated → it invents information or makes claims without sufficient evidence.
- Incomplete → It miss important points.

→ our project AI Response Quality Evaluator Agent a system that checks whether an AI-generated answer is relevant, correct, complete, and free unsupported information.

How will implement it?

1. User input (Question + AI answer + optional reference document)
2. Four Evaluation agents (check relevance, accuracy, hallucinations & completeness)
3. Verdict Agent (combines scores and gives the final Verdict)
4. Dashboard (displays scores, detected issues, and explanations)

DATE: / / .
PAGE NO:

How do the five agents work?

Agent 1:- Relevance judge

→ checks whether the answer actually addresses the user's question.

Ex:- If asked about Python, does the answer discuss Python rather than Java?

Agent 2:- Accuracy judge

→ checks factual correctness by comparing claims against a reference answer or reliable source.

Ex:-

Agent 3:- Hallucination Detection

→ looks for claims that are unsupported by the available evidence or contradict it. A claim being unsupported does not automatically mean it's false.

Agent 4:- Completeness Judge

→ checks whether the answer excludes the important information needed to answer the question.

Agent 5:- Verdict Agent

→ checks the other agents results, calculates the overall score using our scoring rules, and returns Pass, Fail, or Review.

Implementation

Step 1:- Build the interface

→ Use Streamlit to create fields for the user's question, the AI answer, an optional reference answer, and an optional PDF document.

Step 2:- Connect the AI model

→ Use the Gemini API to evaluate the answer. Give each agent a separate prompt, clear evaluation criteria, and a structured output format.

DATE: / /

PAGE NO:

Step 3:- Add reference documents (RAG)

→ extract the text from PDFs, divide it into chunks, convert the chunks into embeddings, and store them in FAISS. Retrieve relevant chunks as evidence for evaluation.

Step 4:- Coordinate the five agents

→ Use LangGraph to manage the workflow: run the four evaluators, pass their results to the Verdict Agent, and handle failures or missing evidence.

Step 5:- Display the results

→ show scores, explanations, detected errors, evidence and the final verdict in the Streamlit dashboard. Add batch evaluation and charts afterward.

How do we know whether our project works?

→ We need to test it using question and answer with known correct results. We can measure:

- Accuracy → how often the system correctly ~~defines~~ identifies correct versus incorrect answers.
- Precision and recall → how reliably it identifies problematic answers and how many actual problems it catches.
- Agent agreement → whether the evaluation results are consistent across repeated tests.
- Latency and cost → how long each evaluation takes how much API usage it requires.

We can also test what happens when no reference document is provided, when the PDF has no relevant information, and when the AI answer contains both correct and incorrect claims.

LLM-as-a-Judge → using an LLM to evaluate another AI's answer.

DATE: / /
PAGE NO:

RAG → giving the evaluator relevant information before it judge factual claims.

Multi-agent workflow → dividing evaluation into specialized agents and combining their results.

Dataset :

Why do we need ?

→ Our system takes a question and an AI-generated answer, evaluates it and gives a Verdict. A dataset gives us examples to test whether those evaluations are correct.

Which dataset should we use ?

2 public datasets + then creating a small custom test dataset of ~~four~~ our complete system.

Dataset 1 :- HaluEval

→ specifically designed to evaluate hallucinations in AI-generated answers, dialogue and summaries. This is the best starting point for testing our hallucination detection agent.

Dataset 2 :- TruthfulQA

→ tests whether AI answers are truthful or reproduce common misconceptions. Useful for checking factual reliability, though it doesn't directly evaluate every one of our five criteria.

Dataset 3 :- Our custom evaluation dataset

→ Create approximately 200-500 labelled examples with questions, AI answer, reference answer, evidence and expected labels for relevance, accuracy, completeness and hallucination. This helps to test all five agents together. The final Verdict Agent uses the expected labels to evaluate whether our overall decisions is reliable.

DATE : / /
PAGE NO :

How will we use the dataset?

The workflow will be:

Load dataset (CSV, JSON or Hugging face dataset)

↓

send examples to our five agents

↓

compare predicted labels with expected labels

↓

Calculate evaluation metrics (Precision, recall, F1-score and accuracy where applicable)